Key numbers

resolution (ROI)	$0.1^\circ \times 0.05^\circ$
beam spacing @ 250 m	44 cm × 22 cm
datasheet point	210 m @ 10 %: 90 %/beam
p per beam @ 250 m	0.80 / 0.67 / 0.21 (clear / rain / fog)
stopping @ 130 km/h	127 m, 68 frames left

250 m, 10 %, clear	n	$E[K]$	R_{99} [m]
pedestrian	9.5	7.6	**235**
motorcycle	11.8	9.4	**237**
car	28.4	22.8	**≥300**
truck	91.9	73.8	**≥300**
tyre / cargo	2.6	2.1	**143**

Pedestrian: $P(K \geq 5) = 96.2\,\%$ clear, 80 % rain, 5 % fog $\rightarrow$ 91 % weighted (88/8/4 %)

Datasheet source: where 90 % at 210 m comes from

InnovizTwo GMSL Series Production Datasheet v3.1 (2024), Condor configuration

- range line: "0.1°x0.05°: 210 m @ 10 % Reflectivity"
- footnote 4: 25 °C; 10 % Lambertian target; 100 klx ambient light; clear weather; no blockage on window; Normal power mode.
  "True Positives = 90 % per pixel and False Positives = 1 % per pixel"
- footnote 1: $0.05^\circ \times 0.05^\circ$ can be configured – trade-off with frame rate, FoV, range, power

Mapping to the model

$$R_0 = 210\,\text{m}, \quad \rho_0 = 0.10, \quad P_0 = 0.90 \quad \Rightarrow \quad p(R_0, \rho_0) = P_0$$

Same convention as lecture 3, slide 20 (Seyond Falcon: 250 m @ 10 %, 100 klx, POD > 90 %)

Geometry: why the beam count is random

Beams at $x_j = u + j\,s$, $s = R\,\Delta\theta$, offset $u \sim U[0, s)$

Object side $[0, W]$, $L = W/s$:

$$n_h = \begin{cases} \lfloor L \rfloor + 1 & \text{if } u < W - \lfloor L \rfloor s \\ \lfloor L \rfloor & \text{otherwise} \end{cases}$$

$$\Rightarrow \; P(n_h = \lfloor L \rfloor + 1) = L - \lfloor L \rfloor, \qquad E[n_h] = L$$

Same vertically, independently.

Pedestrian @ 250 m (0.5×1.8 m)

	horizontal	vertical
spacing s	0.436 m	0.218 m
L	1.15	8.25
counts	1 or 2	8 or 9
probability	85 % / 15 %	75 % / 25 %

$\Rightarrow$ mostly **a single column** of 8–9 beams: the horizontal resolution is the bottleneck

Detection probability per beam

Signal (lidar equation, Lambertian, target > spot): $P_r \propto \dfrac{\rho}{R^2}\, e^{-2\alpha R}$

Detection: signal events $\sim$ Poisson(λ), $\lambda \propto P_r$; return if ≥ 1 event:

$$p = 1 - e^{-\lambda} = 1 - \exp\Big(- C\, \frac{\rho}{R^2}\, e^{-2\alpha R} \Big)$$

Calibration $p(210\,\mathrm{m}, 10\,\%) = 0.9$:

$$C = -\ln(1 - P_0)\, \frac{R_0^2}{\rho_0}$$

At 250 m:
$\ln 10 \cdot (210/250)^2 = 1.625 \Rightarrow p = 0.803$

R [m]	p (10 %)	p (5 %)
150	0.989	0.895
200	0.921	0.719
210	**0.900**	0.684
235	0.841	0.601
250	**0.803**	0.556
275	0.739	0.489
300	0.676	0.431

one free constant, fixed by the one datasheet point

Statistics: worked example (pedestrian, 250 m, 10 %, clear)

$$P(K \geq 5 \mid n) = 1 - \sum_{i=0}^{4} \binom{n}{i} p^i (1-p)^{n-i}, \qquad P(K \geq 5) = \sum_{ij} w_{ij}\, P(K \geq 5 \mid n_{ij})$$

n	weight w	$P(K \geq 5 \mid n)$	contribution
$1 \times 8 = 8$	$0.85 \cdot 0.75 = 0.64$	0.947	0.604
$1 \times 9 = 9$	$0.85 \cdot 0.25 = 0.21$	0.982	0.209
$2 \times 8 = 16$	$0.15 \cdot 0.75 = 0.11$	1.000	0.112
$2 \times 9 = 18$	$0.15 \cdot 0.25 = 0.04$	1.000	0.037
			0.962 < 0.99

$E[K] = 1.15 \cdot 8.25 \cdot 0.803 = 7.6 \geq 5$ but $P(K < 5) = 3.8\,\%$

the average passes, the probability fails

Verification

1. **Calibration:** $p(210\,\text{m}, 10\,\%) = 0.900$ exactly
2. **Hand calculation** (B5) = code output: 0.9618
3. **Monte Carlo**, independent of the formulas:
   200 000 frames: random grid offset $u \sim U[0, s)$, count beams, draw $K \sim \mathrm{Bin}(n, p)$

pedestrian, 250 m	formula	Monte Carlo
$P(K \geq 5)$	96.18 %	96.19 %
$E[K]$	7.60	7.60

Why not Poisson for K? With $p = 0.8$ the binomial is much narrower: Poisson gives 87 % (too pessimistic)

Sensitivity: what changes the pedestrian verdict?

variant	P @ 250 m	R_{99}
our model	**96.2 %**	**235 m**
speckle curve shape	98.9 %	248 m
datasheet $P_0 = 80\,\%$	82 %	214 m
datasheet $P_0 = 95\,\%$	99.3 %	254 m
threshold 95 %	–	253 m
threshold 99.9 %	–	219 m
reflectivity 5 %	59 %	195 m
width 0.4 m	87.6 %	230 m
width 0.6 m	97.2 %	238 m

min. returns k	P @ 250 m	R_{99}
3	99.9 %	274 m
4	99.3 %	254 m
5	**96.2 %**	**235 m**
6	85.7 %	221 m
8	35.2 %	196 m

Robust: cars, trucks pass and the tyre fails under every variant.

Borderline: pedestrian, motorcycle – we say so.

Weather details

Loss on the way out and back: $e^{-2\alpha R}$
Visibility V = distance where contrast drops to 2 %:

$$e^{-\alpha V} = 0.02 \Rightarrow \alpha = \frac{3.912}{V}$$

	V	α [1/m]
clear	> 20 km	0
rain / haze	5 km	$7.8 \cdot 10^{-4}$
fog	1 km	$3.9 \cdot 10^{-3}$

WMO: fog = visibility below 1 km, so 1 km is the mildest fog

Denser fog, 99 % range, 10 %:

V	1 km	500 m	200 m
car	221	175	114
pedestrian	167	144	98
truck	273	208	131

Weights (our estimate), pedestrian weighted P:

clear / rain / fog	P @ 250 m
88 / 8 / 4 %	**91.3 %**
80 / 12 / 8 %	87.0 %
95 / 4 / 1 %	94.6 %

Higher resolution: the $0.05^\circ \times 0.05^\circ$ option

Footnote 1: configurable, with trade-offs in frame rate, FoV, range and power.

If the 210 m @ 10 % spec stayed the same (optimistic):

10 %, clear	R_{99} at 0.1° (ROI)	R_{99} at 0.05°
pedestrian	**235 m**	**290 m**
motorcycle	**237 m**	**$\geq$300 m**
tyre / cargo	**143 m**	**194 m**

$\Rightarrow$ confirms that the **horizontal resolution is the bottleneck**, but the datasheet says range / frame rate pay for it, so we don't claim it.

Last year's InnovizTwo team assumed 0.05° everywhere, judged at 200 m and used averages only.

EBA context: stopping distance and frames

$$d = v\,t_{\text{lat}} + \frac{v^2}{2a}, \quad t_{\text{lat}} = 0.5\,\text{s}, \quad a = 6\,\text{m/s}^2$$

speed	d	time to 250 m
100 km/h	78 m	9.0 s
130 km/h	**127 m**	**6.9 s**
140 km/h	146 m	6.4 s

From 250 m to 127 m: 3.4 s = **68 frames** @ 20 FPS

Several frames? If independent:

frames	P (pedestrian, 250 m)
1	96.2 %
2	99.85 %
3	99.994 %

But frames are correlated (same object, distance, grid offset, weather) → an upper bound only; our verdict uses single frames.